

# CE 310 — Week 3 Assignment: Conditional Probability and Engineering Decisions

Due: Wed Sep 16, 9:00 AM — submit to D2L

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## Overview

This assignment extends the Week 3 in-class exercise into new territory. The exercise computed conditional probabilities directly from the data; here you reverse one — given that an alarm fired, how likely is the fault? — using Bayes' theorem, which the exercise never touches. You then apply the exercise's own conditional results to a sizing decision, and explain in writing why conditioning on context changes what a probability means for design.

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## How to Submit

**Read this before you submit.** Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

| Requirement      | Detail                                                                                                                                                                      |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File format      | .xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF                                                                                          |
| Sheet name       | Must be exactly Answers — do not rename or delete it                                                                                                                        |
| Where answers go | <b>Column A of the Answers sheet labels every row</b> — P1a_PFlag, P1c_written, P2a_written, and so on. Each question names its row; type into the shaded cell on that row  |
| Column layout    | Enter your formula (or, for Problem 3, your arithmetic) in <b>Column C</b> and the resulting value in <b>Column D</b> — do not insert, delete, or move columns              |
| Formulas         | Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90")', so Excel stores the formula instead of running it. <b>Screenshots are not accepted</b> |

| Requirement       | Detail                                                                                                                                                                                                                                                  |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Written responses | Type them in the yellow cells lower down the same sheet — do not submit a separate PDF                                                                                                                                                                  |
| Problem 1 working | Show every intermediate step in the Answer Sheet itself: Steps 1 and 2 in the P1a_PFlag and P1a_posterior rows, the frequency table in the P1b_* rows, the recomputation in P1d_PFlag and P1d_posterior, and typed prose in P1c_written and P1d_written |
| Name and NetID    | Type them in the <b>Name</b> and <b>NetID</b> cells at the top of the Answers sheet — that is what matches your work to your student record. A 5-point deduction applies if the NetID is blank.                                                         |
| File name         | CE310_W3_Assignment_<NetID>.xlsx (e.g. CE310_W3_Assignment_abc123.xlsx)                                                                                                                                                                                 |
| Submission        | Upload the .xlsx file to D2L → <b>Assignments by 9:00 AM, Wed Sep 16</b>                                                                                                                                                                                |

### Problem 1 — Bayes' Theorem: HVAC Fault Detection (30 pts)

A chilled-water plant serving a Tucson office building is monitored by an automated diagnostic system that flags anomalous condenser water delta-T values. Based on commissioning records and three years of maintenance logs:

| Known quantity                                                                        | Value       |
|---------------------------------------------------------------------------------------|-------------|
| P(Fouling) — fraction of summer operating hours with significant chiller tube fouling | <b>0.12</b> |
| P(Flag   Fouling) — probability the diagnostic flags the hour when fouling is present | <b>0.88</b> |
| P(Flag   No Fouling) — false positive rate                                            | <b>0.18</b> |

**Part 1a — Compute P(Fouling | Flag) (12 pts)**

A flag has just fired. What is the probability that the chiller actually has significant fouling?

**Step 1 (4 pts):** Compute P(Flag) using the total probability rule:

$$P(\text{Flag}) = P(\text{Flag} \mid \text{Fouling}) \cdot P(\text{Fouling}) + P(\text{Flag} \mid \text{No Fouling}) \cdot P(\text{No Fouling})$$

**Step 2 (8 pts):** Apply Bayes' theorem:

$$P(\text{Fouling} \mid \text{Flag}) = \frac{P(\text{Flag} \mid \text{Fouling}) \cdot P(\text{Fouling})}{P(\text{Flag})}$$

In your work, label which value is the prior, which is the likelihood, and which is the posterior.

Show all arithmetic. Report your final answer as a percentage rounded to one decimal place.

Enter  $P(\text{Flag})$  in the  $P1a\_PFlag$  row and  $P(\text{Fouling} \mid \text{Flag})$  in the  $P1a\_posterior$  row. Put the arithmetic in Column C and the value in Column D.

**Part 1b — Verify With a Frequency Table (5 pts)**

Imagine **1,000 summer operating hours**. Fill in the table using the given probabilities:

|                 | Flag fires | No flag | Row total |
|-----------------|------------|---------|-----------|
| Fouling present | ?          | ?       | ?         |
| No fouling      | ?          | ?       | ?         |
| Column total    | ?          | ?       | 1,000     |

Compute  $P(\text{Fouling} \mid \text{Flag})$  from the table as (Fouling AND Flag) ÷ (all Flag). Confirm it matches Part 1a within rounding.

Fill in the  $P1b\_fouling$ ,  $P1b\_nofouling$  and  $P1b\_total$  rows. The grand total (1,000) is already entered.

**Note:** Fractional hours are correct here — the question says “imagine 1,000 hours”, so a value like 158.4 is valid. Do not round to whole hours.

### Part 1c — Engineering Interpretation (3 pts, written)

In 2–3 sentences: the diagnostic detects 88% of fouled hours, yet your Part 1a posterior is far lower. Explain why those two numbers are consistent — cite your own Part 1a value — and state what maintenance protocol this suggests: emergency shutdown, scheduled inspection, or something in between?

*Enter your response in the shaded cell on row P1c\_written of the Answers sheet.*

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### Part 1d — Would a Better Sensor Be Worth Buying? (10 pts)

A vendor offers a diagnostic with the same 88% detection rate but a **false-positive rate of 0.05** instead of 0.18. Everything else is unchanged.

**Recompute both quantities with the new false-positive rate (6 pts):**

*Enter them in the P1d\_PFlag and P1d\_posterior rows, arithmetic in Column C.*

**Then, in 2–3 sentences (4 pts):** Is the new sensor worth buying? Cite **both** posteriors. Note which part of the Bayes calculation changed and which did not — one of the two numbers in the fraction is identical to Part 1a. Then say what the better sensor does *not* fix.

*Enter your response in the shaded cell on row P1d\_written of the Answers sheet.*

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## Problem 2 — Cooling System Sizing and Peak Month Demand (18 pts)

Your exercise results for the summer and peak-month cooling picture: -  $P(\text{Cooling} > 50 \text{ kWh} \mid \text{Summer}) = 37.82\%$  (exercise cell B11) - Month with highest  $P(\text{Cooling} > 50) = \text{June}$  (exercise cell B12) - Mean Cooling during Summer + BusinessHours = 74.48 kWh (exercise cell B13)

This problem uses those values to reason about HVAC sizing, then adds one new computation at a higher threshold.

**2a.** ASHRAE 90.1-2022 (Energy Standard for Buildings Except Low-Rise Residential Buildings) requires that cooling systems be sized for peak load conditions — not average seasonal load. Using the exercise values above, explain in 2–3 sentences why mean cooling during summer business hours (74.48 kWh) is an insufficient basis for sizing a cooling plant. What does  $P(\text{Cooling} > 50 \mid \text{Summer}) = 37.82\%$  add to that picture?

Enter your response in the shaded cell on row P2a\_written of the Answers sheet.

**2b.** A demand-response contract prices summer weekday peaks separately, so the utility needs to know how often the building draws more than 150 kWh in an hour during summer weekdays — the threshold above which the peak tariff applies.

Calculate:

**P(Electricity > 150 kWh | Season = Summer AND DayType = Weekday)**

**Note:** This is a two-condition COUNTIFS on Season (column E) and DayType (column D), with the threshold on Electricity (column G).

Show your formula and report the result.

Enter this in the shaded cell on row P2b of the Answers sheet.

**2c.** In 3–4 sentences, synthesize three values into a recommendation for sizing this building's cooling system: your own 2b result, the exercise's mean cooling of 74.48 kWh (B13), and the exercise's P(Cooling > 50 | Summer) = 37.82% (B11). Would sizing the system for exactly the mean load (74.48 kWh) be adequate? Reference ASHRAE 90.1's requirement that cooling systems be sized for peak load conditions, not annual average load. Frame the recommendation around when the peak tariff threshold is crossed.

Enter your response in the shaded cell on row P2c\_written of the Answers sheet.

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### Problem 3 — Reflection (12 pts)

Answer in **one paragraph (100–150 words)**:

A colleague — a project manager with an engineering background — reviews your Week 3 exercise summary and says: “Conditional probability is just a technicality — if 14% of all hours have high cooling demand, I’ll just size the system for 14% load frequency. The conditional breakdowns by season or business hours are interesting but they don’t change the bottom line.”

Using at least **one specific number from your exercise or this assignment**, write a professional response to this claim. Your response should address whether the 14% unconditional figure captures what a building systems engineer actually needs to know for design, and what the conditional values reveal that the marginal probability obscures.

Enter your response in the shaded cell on row P3\_written of the Answers sheet.

## Grading Summary

| Problem                                                                                                                 | Points        |
|-------------------------------------------------------------------------------------------------------------------------|---------------|
| P1a — $P(\text{Flag})$ and $P(\text{Fouling} \mid \text{Flag})$ , arithmetic shown, prior/likelihood/posterior labelled | 12            |
| P1b — Frequency table: eight cells, and the posterior recovered from it                                                 | 5             |
| P1c — Written: why 88% detection and a ~40% posterior are consistent                                                    | 3             |
| P1d — Recompute at a 0.05 false-positive rate (6) + written procurement judgment (4)                                    | 10            |
| P2 — Cooling System Sizing and Peak Month Demand (2a written, 2b computed, 2c written)                                  | 18            |
| P3 — Reflection paragraph (100–150 words, at least one specific number cited)                                           | 12            |
| <b>Total</b>                                                                                                            | <b>60 pts</b> |

This assignment is worth 60 points, the same as every other week. Until August 2026 it was 80: Bayes sat in fourth place behind three conditional-probability computations that the exercise already covers 22 times over. Bayes is now Problem 1.